
IK 509: Research methodology

Course Code	IK 509
Course Name	Research methodology
L-T-P-C	1-0-0-1
Prerequisites	None
Intended for	MTech(R) and PhD students
Mutual Exclusion	None
Approval	53rd BoA (Dec 2023)

Course Contents

- ReseaHii Philosophies and paradigms (1 hour)
 - Six systems of Indian Philosophy- Introduction
 - Research paradigms in Indian philosophical system
 - Eastern & Western paradigms-critical overview
- Research Ethics (1 hour)
 - Ethics from Indian knowledge system perspective (IKS)
 - Evolution of human research ethics
 - Literature review and referencing (1 hour)
 - Review of classical scriptures of Indian philosophy
 - Relevant modern science literature review
- Sampling methods (1 hour)
 - Non-probability sampling methods
 - Probability sampling methods
- Research designs (1 hour)
 - Experimental designs
 - Non-experimental designs
- Clinical trial and control techniques (1 hour)
 - Confounders and control techniques
 - Controlled clinical trials (CCT)
- Logic of hypothesis testing (1 hour)
 - Null hypothesis significance testing (NHST)
 - Steps of hypothesis testing
- Data collection methods (1 hour)
 - Measurement errors and bias
 - First person subjective methods.
 - Third person objective methods
- Procedure for conducting research (1 hour)
 - Ethics committee approval
 - Clinical trial registration
 - Informed consent assent
 - Pilot study- need and importance
 - Protocol- apriori design vs. interim modifications

- Managing adverse effects
- Fundamentals of qualitative research (1 hour)
 - Major qualitative research approaches
 - Validity & reliability in qualitative research Technical software for Research scholars (1 hour)
 - Translator- for Sanskrit transliteration
 - R & Jamovi-for statistical analysis
 - Zotero-for referencing
- Statistics over view (1 hour)
 - Descriptive statistics
 - Fundamentals of inferential statistics
- Statistics overview-2 (1 hour)
 - Choosing the right statistical tests
- Scientific writing (1 hour)
 - Classical scripmres perspective
 - Modern science perspective

Text Books

1. Aron A, Aron EN, **Statistics for psychology**, 6th Edition, Pearson Education, 2013.
2. Gough AE, Cowell EB, **The Sarva-Darsana-Samgraha**: Or, Review of the Different Systems of Hindu Philosophy. DigiCat; 2022.

References

1. Zar JH, **Biostatistical analysis**, Pearson Education India; 1999.
2. Crewell JW, Poth CN, **Qualitative inquiry and research design: Choosing among fixe approaches**, Sage publications, 2016
3. Radhakrishnan S, Moore CA, (editors), **A source book in Indian philosophy**, Princeton University Press, 1957.